Lesson notes · for you

Functions: one job each

Programming · Level 3 · Lesson 1

Time	Materials
20 min	A phone with this lesson downloaded; A notebook for tests

Learning goals

- Define functions with parameters and return a result.
- Tell return from print.
- Test a function with assert.

Activity: Function machines

1. Warm-up (3 min): A partner thinks of a rule (“double it and add 1”). Give inputs; they answer with outputs. Guess the rule: that is a function.
2. Write (7 min): Write the rule as def f(n): return ..., plus three asserts, one of them an edge case such as 0 or a negative number.
3. Break (7 min): Swap with your partner. Write one new assert that you think their function will fail. Run it.
4. Fix (3 min): If your function broke, fix it, then run all the asserts again.

The warm-up works with no phone at all: input in, output out.

Discussion

- Why is return more useful than print inside a function?
- Which edge case broke a function today?
- How do you choose a good name for a function?

Check yourself

- Write a function that returns the bigger of two numbers, and one assert for it.

Try it this week

Turn one repeated piece of a program you wrote into a function, and write two asserts for it before you use it.

Worksheet

Functions: one job each

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Name _____ Class _____ Date _____

1 **You wrote average(marks), and it passes a test with three marks. Which extra test is most useful?**

◯ An empty list ◯ Another easy list of two marks ◯ The same easy test again

2 **Write the function average(marks): it returns the average of the marks, and 0 for an empty list. The tests call it with asserts.**

```
def average(marks):
    # average; 0 for an empty list
    pass
```

Write your program:

3 **What does this program print?**

```
def double(n):
    print(n * 2)

x = double(5)
print(x)
```

◯ None, then 10 ◯ 10, then 10 ◯ Only 10 ◯ 10, then None

4 **Put the lines in order: define area, call it for a 4 by 3 room, and print Area: 12. One line doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `return length * width`

☐ `room = area(4, 3)`

☐ `def area(length, width):`

☐ `print(length * width)`

☐ `print("Area:", room)`

5 **Variables created inside a function can still be used after the function has finished.**

◯ True ◯ False

✎ **Look at a program you have written. Which lines repeat, or do one clear job? Name the function you would make, its parameters, and what it returns.**

Check your answers

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1 **An empty list**

The empty list is the edge case: it divides by zero, so you must decide what average([]) should return.

2

```
def average(marks):
    if len(marks) == 0:
        return 0
    return sum(marks) / len(marks)
```

The function returns its result so the tests (and other code) can use it.

3 **10, then None**

double prints 10 while it runs, but has no return, so x is None and the second print shows None.

4 **Top to bottom: 2, 3, 1, ✗, 4**

```
def area(length, width):
    return length * width
room = area(4, 3)
print("Area:", room)
```

The function must be defined before it is called, and it must return the result for room to hold 12.

5 **False**

A function's variables are its own and disappear when it returns.